Embedding Space Analysis for Heterogeneous Graph Data: A Study in the Context of Wild-GAD

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ABSTRACT

Graph Anomaly Detection (GAD) is gaining increasing relevance in practical domains such as financial graphs, social networks, and knowledge graphs. However, the development of effective GAD systems faces several fundamental challenges, as graph data is often highly heterogeneous with respect to both structure and domain.

To address this challenge, Cao et al. [2] propose the *Wild-GAD* framework, which leverages pretrained language models to project textual node features into a shared semantic space and employs a dataset selection mechanism to incorporate external graph data from heterogeneous domains.

The objective of this report is to analyze the representation-level assumptions underlying this approach, evaluate alternative language models, and study how heterogeneous graph datasets can be embedded within a shared feature space.

ACM Reference Format:
Tristan Wesendahl Velázquez. 2026. Embedding Space Analysis for Heterogeneous Graph Data: A Study in the Context of Wild-GAD. In *Papers/Projects of The Knowledge-Based Systems Seminar 2025/2026, UOS, Germany.* 9 pages.

1 INTRODUCTION

According to Cao et al. [2], projecting textual node features using mBERT[7] leads to several notable effects. In particular, this projection improves comparability across heterogeneous data domains, results in more stable clusters within a shared feature space, and reduces domain gaps compared to traditional bag-of-words representations.

1.1 The Wild-GAD Framework

The experimental results demonstrate that external graph data can serve as valuable reference domains for graph anomaly detection. The Dataset Selector module is shown to be capable of identifying suitable external graphs whose inclusion improves anomaly detection performance. Furthermore, the use of augmented datasets often yields more consistent results than their original counterparts, suggesting that augmentation can act as a stabilizing factor in cross-domain settings.

In addition, the results indicate that **Wild-GAD(M)**, which incorporates two selected external graphs, frequently outperforms **Wild-GAD(S)**, which relies on only a single external graph. This suggests that leveraging multiple reference domains can provide richer contextual information than a single reference alone. Interestingly, the inclusion of labeled external data does not consistently improve performance and may even degrade results in certain domains.

Overall, the dataset selection mechanism proves effective in identifying beneficial external reference graphs.

When compared to semi-supervised graph anomaly detection methods with a high proportion of labeled data, such as BernNet [10], AMNet [3], BWGNN [20], GATSep [25], DCI [23], and GHRN [8], Wild-GAD [2] consistently achieves superior performance in terms of both AUROC and AUCPR. These results highlight the strength of the framework, particularly its ability to achieve competitive or superior performance without relying on label information.

Despite these promising results, the paper also identifies several limitations. Most notably, the framework exhibits a strong dependence on mBERT[7] as the underlying text encoder, and the results vary considerably across different data domains, indicating limited robustness to domain-specific characteristics.

1.2 Motivation

The Wild-GAD [2] framework demonstrates that language model embeddings and external reference graphs can substantially improve the applicability of graph anomaly detection methods to heterogeneous, real-world data. The reported results indicate improved cross-domain comparability, more stable clustering behavior, and competitive performance even without relying on labeled data. These findings highlight the potential of embedding-based representations as a foundation for using graph data from the wild.

At the same time, the paper leaves several questions open that are directly relevant to the design of embedding-based GAD pipelines. Most notably, the framework relies exclusively on mBERT[7] as the underlying text encoder, while alternative language models are not considered. As a result, it remains unclear whether the observed improvements are primarily due to the general framework design or to properties specific to mBERT[7].

Furthermore, the paper does not analyze how different language models shape the geometry of the embedding space across heterogeneous domains. In particular, it is unclear to what extent embeddings capture shared semantic structure versus domain-specific characteristics, and how this balance affects cross-domain comparability.

Motivated by these open questions, this seminar paper focuses on analyzing embedding spaces induced by different language models in the context of Wild-GAD [2]. The goal is to investigate whether alternative language models yield embedding geometries that are more suitable for heterogeneous graph data, thereby providing deeper insight into the representation-level foundations of embedding-based graph anomaly detection. In particular, we explicitly analyze whether

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Wild-GAD's dataset[2] selection mechanism yields invariant selections across embedding models, or whether selection outcomes change as a function of the induced embedding geometry.

Since the dataset selection mechanism relies entirely on embedding-level similarity, the choice of the underlying language model implicitly defines the geometry of the selection space. The following sections, therefore, investigate how different language models shape this feature space and how this affects dataset comparability as well as the dataset selection of the Wild-GAD Framework[2].

2 RELATED WORK

2.1 Dataset Selection in Wild-GAD

The Wild-GAD framework [2] is designed to leverage external graph datasets in order to improve unsupervised graph anomaly detection under data-scarce conditions. At its core, Wild-GAD [2] relies on a dataset selection mechanism that operates entirely at the representation level and is independent of the downstream anomaly detection model.

Unified Embedding Space. Given a target graph G_t and a collection of candidate external graphs $\{G_1, \ldots, G_m\}$, Wild-GAD [2] first maps all textual node attributes into a shared semantic embedding space using a fixed pretrained language model. Let $z_i \in \mathbb{R}^d$ denote the embedding of node v_i, obtained by applying the language model encoder without task-specific fine-tuning. This step enables heterogeneous graph datasets originating from different domains to be compared within a common feature space.

In the original Wild-GAD framework [2], **multilingual BERT (mBERT)** [7] is used as the embedding model, thereby implicitly defining the geometry of the selection space.

Spherical Representation of the Target Graph. After obtaining node embeddings for the target graph G_t, Wild-GAD [2] characterizes the embedding distribution of normal nodes using a spherical coordinate representation centered at the mean embedding. Specifically, the center of the target embedding space is defined as

$$\mathbf{c} = \frac{1}{|V_t|} \sum_{v_i \in V_t} z_i,$$

where V_t denotes the node set of the target graph. Each node embedding is then described by its radial distance $r_i = \|z_i - \mathbf{c}\|$ and its normalized direction vector

$$\boldsymbol{\phi}_i = \frac{z_i - \mathbf{c}}{\|z_i - \mathbf{c}\|}.$$

Representativity Criterion. To assess whether an external dataset is suitable as reference data, Wild-GAD [2] introduces a representativity criterion that measures how strongly the embedding distribution of the target graph is affected by the inclusion of external data. This criterion consists of a center-shift term and a distributional similarity term, where the latter is computed using the Wasserstein distance between spherical coordinate distributions.

Diversity Criterion. In addition to representativity, Wild-GAD [2] explicitly encourages diversity among selected datasets. This is achieved by minimizing the Minimum Hyperspherical Energy (MHE), which favors a uniform coverage of the embedding space and discourages redundant reference datasets. In particular, this term penalizes the selection of multiple external datasets that occupy similar regions of the embedding space.

Final Selection Objective. The representativity and diversity scores are combined into a single objective function

$$J(G) = \eta \cdot \text{mean}(\hat{s}_{\text{center}} + \hat{s}_{\text{dist}}) + (1 - \eta)\hat{s}_{\text{div}},$$

where $\eta \in [0, 1]$ controls the trade-off between similarity and diversity. The datasets with the lowest objective values are selected as external reference graphs, with the number of selected datasets determined by the Wild-GAD(S) or Wild-GAD(M) configuration [2].

Motivation for Model Analysis. Since the dataset selection process operates entirely in the embedding space, the choice of the underlying language model directly influences dataset comparison and ranking. In the original Wild-GAD framework [2], mBERT [7] serves as the default embedding model and thus implicitly defines the selection geometry. The present seminar paper builds upon this observation by analyzing how alternative language models shape the structure of the embedding space and, consequently, affect embedding-based dataset selection behavior.

2.2 Considered Language Models

For the experimental analysis, a set of transformer-based language models is considered, covering different representation paradigms, training objectives, and domain specializations. All models are used to generate fixed-size vector representations of textual node features, enabling embedding-based comparisons across datasets.

Multilingual Models. **Multilingual BERT** [7](mBERT) and **XLM-RoBERTa**[5] (XLM-R) are transformer-based language models trained on large-scale multilingual corpora. These models are designed to produce contextualized representations across a wide range of languages and are commonly used in cross-lingual and multilingual natural language processing tasks. mBERT[7] is introduced by Devlin et al., while XLM-R[5] extends this paradigm using larger multilingual corpora and optimized training strategies.

Sentence Embedding Models. **Sentence-BERT**[18] (SBERT) models are trained using contrastive learning objectives to produce sentence-level embeddings that can be directly compared using vector similarity measures. In contrast to standard BERT[7] models, SBERT[18] outputs fixed-size representations optimized for semantic similarity, clustering, and retrieval tasks. The SBERT[18] framework is proposed by Reimers and Gurevych and forms the basis for variants such as **MiniLM**[22] and **MPNet**[19]. In the experimental analysis, we employ two SBERT-based model variants, namely `all-MiniLM-L6-v2` (SBERT-MiniLM) (SBERT with MiniLM backbone) [18, 22] and `allenai-specter` (SBERT-SPECTER) (SPECTER-based SBERT variant) [4, 18] , which instantiate the Sentence-BERT framework [18] using MiniLM [22] and SPECTER [4] backbones, respectively.

Domain-Specific Models. **SciBERT**[1] is a language model pretrained on large-scale scientific text corpora and is tailored to academic writing and terminology. **SPECTER**[4] further extends this approach by incorporating citation graph information during training, enabling the generation of document embeddings that capture semantic relatedness within citation networks.

Retrieval-Oriented Models. The **e5 model family**[21] is trained using large-scale contrastive learning on information retrieval datasets and is designed to produce embeddings suitable for similarity search and retrieval scenarios. These models support instruction-style input formats, such as query and passage representations, and are introduced by Wang et al. [21].

Monolingual Models. **RoBERTa**[13] is a transformer-based language model trained on large-scale English text corpora using an optimized masked language modeling objective. It serves as a representative monolingual encoder producing contextualized textual representations.

While several representative language models are discussed for contextualization, only a subset of these models is used in the experimental evaluation, as specified in Section 3.

2.3 Graph Benchmark Datasets across Domains and their Characteristics

Graph-based learning methods are commonly evaluated on a diverse set of benchmark datasets originating from multiple application domains. These benchmarks differ substantially with respect to graph structure, textual characteristics, and interaction semantics, reflecting the heterogeneity of real-world graph data.

Citation-Based Graphs. Citation networks represent scientific publications as nodes connected through citation links. Well-established benchmarks in this category include **Cora**, **CiteSeer**, and **PubMed** [15, 17]. The textual content of these datasets typically consists of abstracts or full papers written in formal and technical language, often characterized by longer documents and specialized terminology. Structurally, citation graphs encode directed relationships between documents, resulting in characteristic sparsity patterns and hierarchical connectivity. Such datasets are widely used for evaluating document classification, link prediction, and representation learning in scholarly domains.

Web- and Hyperlink-Based Graphs. Web-based graph benchmarks model relationships between webpages or online documents via hyperlinks. Classical datasets in this category include **WebKB** [6] and Wikipedia-derived graphs such as **WikiCS** [16]. These datasets are characterized by heterogeneous and often noisy textual content, including boilerplate elements, metadata fragments, HTML-like structures, and varying document lengths. Their graph structures reflect hyperlink navigation patterns rather than semantic similarity alone, resulting in distinct connectivity patterns and graph topologies. Only basic preprocessing is typically applied, which may further contribute to variability in textual representations.

Review-Based Graphs/ e-Commerce Graphs. Review-based benchmarks capture interactions between users, items, and textual reviews on online platforms. Prominent examples include the **Yelp** dataset [24] and the **Amazon review graph family** [14]. The Amazon datasets comprise multiple category-specific subsets, such as *Amazon Computers*, *Amazon Electronics*, *Amazon Photo*, and *Amazon Books*, which share a common interaction structure while differing in domain-specific vocabulary. The textual content in these datasets is typically informal, subjective, and highly variable, reflecting user-generated language and diverse writing styles.

Social Network Graphs. Social network benchmarks model interactions and discussions between users in online communities. Datasets such as **Reddit** [9] and Twitter [12]-based graphs represent communication patterns, comment structures, or follower relationships, often combined with short and conversational text. These datasets are characterized by dynamic graph structures, high linguistic variability, and rapidly evolving interaction patterns.

Summary. Across these domains, graph benchmark datasets exhibit substantial variation in both linguistic and structural properties. Differences in document length, vocabulary, formality, interaction semantics, and graph topology pose significant challenges for representation learning and cross-domain comparison. This diversity motivates the need for robust embedding models and principled dataset selection strategies, particularly when transferring information across domains, as addressed in frameworks such as Wild-GAD [2].

3 EXPERIMENTAL SETUP

3.1 Overview

This section describes the experimental setup used to analyze feature spaces induced by different language models in the context of heterogeneous graph data. The experiments are designed to study representation-level properties rather than to evaluate graph anomaly detection performance. The experimental analysis focuses on comparing multiple language models with respect to their ability to project heterogeneous textual node features into a shared embedding space. In addition to representation-level distance analysis, we study how these embeddings influence dataset selection within the Wild-GAD framework[2]. To this end, embeddings are generated for multiple datasets spanning different application domains and analyzed using distance-based metrics and low-dimensional visualizations.

3.2 Embedding Models

The selection of embedding models is a central design choice of this work, as the objective is to analyze how different embedding paradigms induce distinct geometric structures in the resulting representation spaces. Rather than comparing a large number of closely related models, we deliberately select a small but diverse set of language models that represent fundamentally different design principles.

To this end, the considered models are grouped along three key dimensions: (i) the training objective and level of sentence-level optimization, (ii) domain specialization versus general-purpose pretraining, and (iii) architectural backbone and model capacity.

Sentence-Level Embedding Models. A first category comprises models explicitly optimized for sentence-level semantic similarity. These models are particularly relevant for embedding-based distance analyses, as their training objectives directly shape cosine-based geometry. From this category, we select two representative Sentence-BERT variants [18]: `all-MiniLM-L6-v2`, a lightweight SBERT model using a MiniLM backbone [22], and `allenai-specter`, which integrates citation-informed pretraining through the SPECTER

model [4]. This choice allows us to contrast a general-purpose sentence embedding model with a domain-informed variant tailored to scientific text.

General-Purpose Multilingual and Monolingual Models. To provide a broader reference point, we include general-purpose transformer models that are not explicitly optimized for sentence embeddings. Multilingual BERT (mBERT) [7] serves as a widely used baseline for multilingual text representations and is also employed as the embedding model in the original Wild-GAD framework [2]. Including mBERT therefore establishes a direct point of reference to the Wild-GAD selection mechanism and enables a consistent comparison between our embedding space analysis and the dataset selection behavior observed in prior work. RoBERTa [13] is included as a strong monolingual counterpart with optimized pretraining dynamics, allowing us to contrast multilingual and monolingual representations under otherwise comparable settings.

Domain-Specific and Contrastively Trained Models. Finally, we include models designed to incorporate either domain-specific knowledge or contrastive training signals. SciBERT [1] captures scientific language more accurately through domain-adaptive pretraining, while the E5-small model [21] represents a family of contrastively trained embedding models optimized for retrieval and semantic similarity tasks across diverse domains. Including these models allows us to assess the impact of domain specialization and contrastive objectives on embedding dispersion and cross-dataset separation.

Final Model Set. Based on these criteria, the experimental analysis considers the following models:

- SBERT with MiniLM backbone (`all-MiniLM-L6-v2`),
- SBERT with SPECTER backbone (`allenai-specter`),
- E5-small,
- mBERT,
- SciBERT, and
- RoBERTa.

Together, this selection covers a broad spectrum of embedding paradigms while remaining compact enough to enable a focused and interpretable analysis. All models are used as fixed feature extractors without task-specific fine-tuning in order to isolate the intrinsic geometric properties induced by their respective pretraining objectives.

3.3 Datasets

The experimental analysis is based on the following datasets:

- **Cora** [15] and **PubMed** [17] are citation-based datasets consisting of scientific publications connected through citation links.
- **WikiCS** [16] and **WebKB** [6] represent web-based and hyperlink-driven graph structures derived from webpages or online documents.
- **Reddit2** [9], a social media dataset capturing interactions and discussions between users.
- **Yelp** [24] and **Amazon_Computers** [14] are review-based datasets containing user-generated textual content associated with products or services.

These datasets cover a broad range of application domains, including scientific communication, web content, social interaction, and user reviews. The domain categorization used in this work follows the high-level domain grouping introduced in the Wild-GAD framework [2], covering citation, review, social, and web-based graph data. The datasets are used for embedding distribution analysis and as downstream and candidate datasets in the Wild-GAD selection procedure[2].

3.4 Analysis of Dataset Selection Behavior

Beyond analyzing embedding space geometry, we investigate how different language models affect the dataset selection mechanism of Wild-GAD[2]. The objective is to determine whether dataset selection is stable across embedding models and sampling budgets or whether it exhibits model-specific biases.

We consider a common pool of all datasets for both downstream tasks and candidate datasets. For large-scale graphs, uniformly sampled subsets are used to ensure computational feasibility. Self-selection is explicitly excluded; i.e., a downstream dataset is never considered as its own candidate. For each downstream dataset and embedding model, candidate datasets are ranked using the original Wild-GAD[2] unlabeled selection criterion. To assess robustness with respect to sampling variability, all dataset selection experiments were conducted under two different maximum node sampling budgets (3000 and 4000), while keeping the random seed and all other parameters fixed. The second sampling budget of 4000 nodes was chosen to represent a moderate increase in sample size while remaining computationally comparable to the 3000-node setting used in prior experiments. We analyze Top-1 selections across all configurations, resulting in 42 (downstream, model) pairs.

3.5 Embedding Generation and Normalization

For each dataset, textual node features are encoded using the respective language model to obtain fixed-dimensional embeddings. Token-level representations are aggregated into a single embedding vector per node using mean pooling, following standard practice for Transformer-based language models.

In the subsequent analyses, cosine-based similarity measures are employed. Although cosine similarity is invariant to vector magnitude, we apply L2-normalization to all embeddings to explicitly enforce a hyperspherical embedding geometry and to restrict the representation to directional information only. This normalization ensures a consistent geometric interpretation across language models and datasets and aligns with the spherical embedding assumptions underlying the Wild-GAD selection mechanism.

We note that L2-normalization removes magnitude-related information, which limits the interpretability of absolute Euclidean distances. Consequently, Euclidean distance-based analyses are considered only in a complementary manner and are interpreted relative to this geometric constraint.

3.6 Distance Computation

Pairwise cosine distances are used to quantify similarity between embeddings, as commonly done in embedding-based semantic similarity analysis [18]. Due to the potentially large number of node pairs,

distances are estimated using random sampling to ensure scalability while preserving representative distance statistics.

3.7 Quantitative Distance Measures

Based on the L2-normalized embeddings described in Section 3.5 and the cosine distance computation introduced in Section 3.6, several distance-based measures are employed to analyze the geometric structure of the embedding distribution induced by different language models.

Intra-dataset distance. The intra-dataset distance quantifies the internal compactness of a dataset by measuring the average cosine distance between embeddings of nodes sampled from the same dataset. Formally, for a dataset D it is defined as

$$d_{\text{intra}}(D) = \mathbb{E}_{i \neq j \sim D}\left[1 - \tilde{\mathbf{e}}_i^\top \tilde{\mathbf{e}}_j\right],$$

where $\tilde{\mathbf{e}}_i$ denotes the ℓ_2-normalized embedding of node i. In practice, this expectation is approximated via random sampling of node pairs to ensure scalability for large graphs.

Inter-dataset distance. The inter-dataset distance measures the separation between two datasets D_a and D_b embedded by the same language model. It is defined as the expected cosine distance between embeddings of nodes sampled across the two datasets:

$$d_{\text{inter}}(D_a, D_b) = \mathbb{E}_{i \sim D_a,\, j \sim D_b}\left[1 - \tilde{\mathbf{e}}_i^\top \tilde{\mathbf{e}}_j\right].$$

Normalized distance ratio. Since different datasets may exhibit varying degrees of internal dispersion, inter-dataset distances are normalized by the corresponding intra-dataset distances. For a dataset pair (D_a, D_b), the normalized distance ratio is defined as

$$r(D_a, D_b) = \frac{d_{\text{inter}}(D_a, D_b)}{\frac{1}{2}\left(d_{\text{intra}}(D_a) + d_{\text{intra}}(D_b)\right)}.$$

This ratio captures relative separation between datasets with respect to their internal embedding compactness, rather than relying on absolute distance values.

Domain-aware analysis and separation score. Each dataset is assigned to one of the four domains introduced in Subsection 2.3 (Citation, Social, Co-purchase, and Hyper-Link). Based on this categorization, dataset pairs are grouped into *same-domain* and *cross-domain* pairs. To quantify domain-level separation, a separation score is defined as the ratio between the mean normalized distance of cross-domain pairs and the mean normalized distance of same-domain pairs:

$$\text{Sep} = \frac{\mathbb{E}_{(D_a,D_b)\in\text{cross}}\left[r(D_a, D_b)\right]}{\max\left(\mathbb{E}_{(D_a,D_b)\in\text{same}}\left[r(D_a, D_b)\right], \varepsilon\right)}, \quad \varepsilon > 0.$$

The small constant ε ensures numerical stability in cases where same-domain distances approach zero due to strong intra-domain compactness. Higher separation scores indicate stronger relative separation between datasets from different domains compared to datasets within the same domain.

3.8 Dimensionality Reduction for Visualization

To provide qualitative insights into the structure of latent spaces, Principal Component Analysis (PCA) is applied for dimensionality reduction [11] to a randomly selected subset of embeddings for each dataset. The resulting two-dimensional projections are used exclusively for visualization purposes and complement the quantitative distance analysis.

4 RESULTS

4.1 Dataset Selection Behavior in Wild-GAD

Tables 1 and 2 jointly illustrate that dataset selection in Wild-GAD is stable at a coarse level, while still exhibiting model-dependent variability. For most downstream datasets, a small number of candidate datasets dominates Top-1 selections across embedding models, indicating persistent high-level similarities. At the same time, the absence of unanimous agreement highlights the influence of embedding-specific inductive biases. Across both sampling budgets of 3000 and 4000, Top-1 dataset selections remained identical in 86% of all (downstream, model) configurations.

Table 1: Top-1 dataset selected by Wild-GAD for each downstream dataset and language model with sampling size of 4000.

Downstream	E5	mBERT	RoBERTa	MiniLM	SPECTER	SciBERT
Amazon	cora	pubmed	cora	cora	cora	cora
Cora	pubmed	amazon	amazon	pubmed	reddit	amazon
PubMed	cora	reddit	reddit	reddit	yelp	reddit
Reddit	pubmed	pubmed	pubmed	pubmed	pubmed	pubmed
WebKB	cora	cora	cora	cora	cora	cora
WikiCS	cora	pubmed	pubmed	pubmed	yelp	yelp
Yelp	pubmed	wikics	wikics	pubmed	pubmed	wikics

Table 2: Top-1 dataset selection frequency across embedding models. For each downstream dataset, the table reports the most frequently selected candidate dataset (mode) and the number of embedding models for which it was selected as Top-1 with sampling size of 4000.

Downstream Dataset	Most Frequent Top-1 Candidate	Frequency
Amazon-Computers	PubMed	4 / 6
Cora	Amazon-Computers	3 / 6
PubMed	Reddit2	5 / 6
Reddit2	PubMed	6 / 6
WebKB	Cora	5 / 6
WikiCS	PubMed	4 / 6
Yelp	PubMed	3 / 6

4.2 Embedding Space Analysis

Table 3: Quantitative comparison of embedding spaces across language models. Reported values are averaged over all dataset pairs using cosine distance. Higher values indicate stronger separation between datasets.

Model	Mean Intra	Mean Inter	Mean Ratio	Sep. Score
SBERT-MiniLM	0.1590	0.3146	1.98	0.99
SBERT-SPECTER	0.0219	0.0785	3.59	0.78
E5-small	0.0286	0.0629	2.20	0.95
mBERT	0.0326	0.1446	4.44	0.78
SciBERT	0.0350	0.1800	5.14	0.71
RoBERTa	0.0134	0.0684	5.10	0.72

Table 4: Average inter-dataset distance ratios for same-domain and cross-domain dataset pairs.

Model	Same-Domain Ratio	Cross-Domain Ratio
SBERT-MiniLM	2.00	1.98
SBERT-SPECTER	4.40	3.45
E5-small	2.30	2.18
mBERT	5.49	4.26
SciBERT	6.82	4.86
RoBERTa	6.68	4.83

Table 3 summarizes the average intra-dataset distances, inter-dataset distances, and the resulting distance ratios for all considered models.

The results reveal substantial differences between language models. Sentence-level models such as SBERT-MiniLM and E5-small exhibit moderate distance ratios, indicating a balanced relationship between dataset separation and intra-dataset compactness. In contrast, models such as SciBERT, RoBERTa, and mBERT[7] yield considerably higher distance ratios, suggesting stronger relative separation between datasets in the induced latent space.

To further analyze domain effects, Table 4 reports average distance ratios for same-domain and cross-domain dataset pairs. Across all evaluated models, same-domain pairs consistently show higher normalized distance ratios than cross-domain pairs. This indicates stronger relative separation under intra-dataset normalization, which can be attributed to lower internal dispersion within same-domain datasets.

Notably, the magnitude of this effect varies across language models. SBERT-MiniLM exhibits nearly identical ratios for same-domain and cross-domain pairs, suggesting weak domain-aware structuring. In contrast, models such as SciBERT and RoBERTa display a clearer differentiation between domain categories, reflecting stronger domain-dependent embedding geometries.

Overall, the quantitative results demonstrate that different language models induce markedly different embedding geometries, influencing both relative dataset separation and the extent to which domain structure is reflected in the embedding space.

4.3 Embedding Space Visualization

Figure 1 presents the resulting PCA projections, with one visualization per language model. Each plot, therefore, illustrates the relative arrangement of datasets within the latent space induced by a specific model, rather than enabling direct comparisons across different models.

The visualizations reveal pronounced differences in the geometric structure of the embedding distribution across models. Sentence-level models such as *SBERT-MiniLM*[18, 22] and *E5-small*[21] exhibit comparatively smooth embedding geometries with compact and partially overlapping dataset clusters, indicating weaker global separation across heterogeneous domains. In contrast, models such as *mBERT*[7], *SciBERT*[1], and *RoBERTa*[13] induce embedding distributions with stronger global separation, in which individual datasets often form clearly distinguishable regions.

These qualitative patterns are broadly consistent with the quantitative distance ratios reported in Table 3 and Table 4. In particular, models that exhibit higher normalized distance ratios tend to produce embedding spaces with stronger apparent separation between datasets, while models with lower ratios show increased overlap in the low-dimensional projections. The PCA visualizations in Figure 1 qualitatively support the quantitative findings. Models with higher separation ratios induce embedding spaces with more clearly separated dataset regions, whereas models with lower ratios show increased overlap.

It should be considered that PCA projections provide only a two-dimensional approximation of the original high-dimensional embedding spaces and primarily serve as a qualitative analysis tool[11]. As a result, relative separation effects captured by normalized distance metrics may not always be fully reflected in the visualizations, and direct visual comparisons across different language models should therefore be interpreted with caution.

5 DISCUSSION

The results show that the choice of language model has a substantial impact on the geometry of the induced embedding distribution and, consequently, on embedding-based similarity judgments between datasets. Differences in mean distance ratios indicate that language models vary considerably in how strongly they separate datasets in a shared representation space. Models with higher ratios induce stronger global separation, while models with lower ratios produce more compact and overlapping embeddings.

5.1 Effects of Language Model Choice

The observed differences between models suggest that pretraining objectives and domain specialization play an important role in shaping embedding geometry. Models pretrained on scientific text, such as SciBERT and SBERT-SPECTER, exhibit increased separation for citation-based datasets, reflecting their specialization in scientific language. In contrast, sentence-level models optimized for general semantic similarity, such as SBERT-MiniLM and E5-small, tend to produce more balanced embedding spaces across domains.

These findings indicate that embedding distribution are not neutral representations but instead reflect inductive biases introduced during pretraining.

5.2 Stability and Implications of Dataset Selection

The results indicate that dataset selection within the Wild-GAD framework [2] is stable at a coarse level, while still exhibiting systematic variability across language models. For most downstream datasets, a small subset of candidate datasets dominates the Top-1 selections across multiple embedding models, suggesting that the selection mechanism captures persistent high-level similarities between datasets.

At the same time, Top-1 selections are not invariant with respect to the choice of language model. Several downstream datasets admit multiple Top-1 candidates depending on the embedding model, reflecting differences in semantic inductive biases induced during language model pretraining. These effects are particularly pronounced for datasets with heterogeneous or multi-domain textual content.

Across sampling budgets, dataset selection remains robust: Top-1 selections are preserved in the majority of cases when increasing the maximum number of sampled nodes from 3000 to 4000. Observed

changes are largely confined to near-tie situations, where small sampling-induced perturbations can alter the ranking order without indicating a substantial shift in dataset similarity. This suggests that the selection mechanism is not overly sensitive to moderate sampling variability.

Overall, these findings imply that variability in dataset selection primarily reflects differences in embedding geometry introduced by the language model, rather than stochastic instability of the selection procedure itself. Consequently, the choice of language model constitutes an implicit but critical design decision in embedding-based dataset selection pipelines within Wild-GAD.

5.3 Limitations

Several limitations of the present analysis should be noted. First, the evaluation is restricted to cosine distance and PCA-based visualization, which may not fully capture higher-dimensional geometric properties of the embedding spaces. Second, the analysis focuses exclusively on representation geometry characteristics and dataset selection behavior, without re-evaluating graph augmentations or anomaly detection performance. Future work could extend this analysis by incorporating additional similarity measures, higher-dimensional analysis techniques, and downstream task evaluations.

6 CONCLUSIONS

In this work, we analyzed how different language models shape embedding spaces for heterogeneous graph data and how these representation-level differences affect embedding-based dataset comparison and selection within the Wild-GAD framework[2]. Using a combination of quantitative distance metrics and qualitative visualizations, we demonstrated that language models induce substantially different embedding geometries, influencing both relative dataset separation and domain-dependent structure.

Beyond embedding space analysis, we explicitly examined the dataset selection behavior of Wild-GAD[2] under varying embedding models and sampling budgets. Our results show that dataset selection is generally robust with respect to moderate sampling variation, while remaining sensitive to the semantic inductive biases introduced by different language models. Selection switches predominantly occur in near-tie situations, indicating fine-grained ambiguity rather than instability of the selection mechanism itself.

These findings highlight that the choice of language model constitutes an implicit but influential design decision in embedding-based graph selection frameworks. While the original use of mBERT in Wild-GAD[2] represents a pragmatic and broadly applicable choice, our analysis suggests that alternative models with stronger domain specialization or improved sentence-level representations may yield different and potentially more suitable selection outcomes depending on the application context.

Overall, this work underscores the importance of explicitly considering representation geometry effects when designing and interpreting embedding-based dataset selection pipelines. Future work may extend this analysis by integrating structural graph features, exploring alternative similarity measures, or evaluating the downstream impact of selection variability on anomaly detection performance.

A EMBEDDING SPACE VISUALIZATIONS

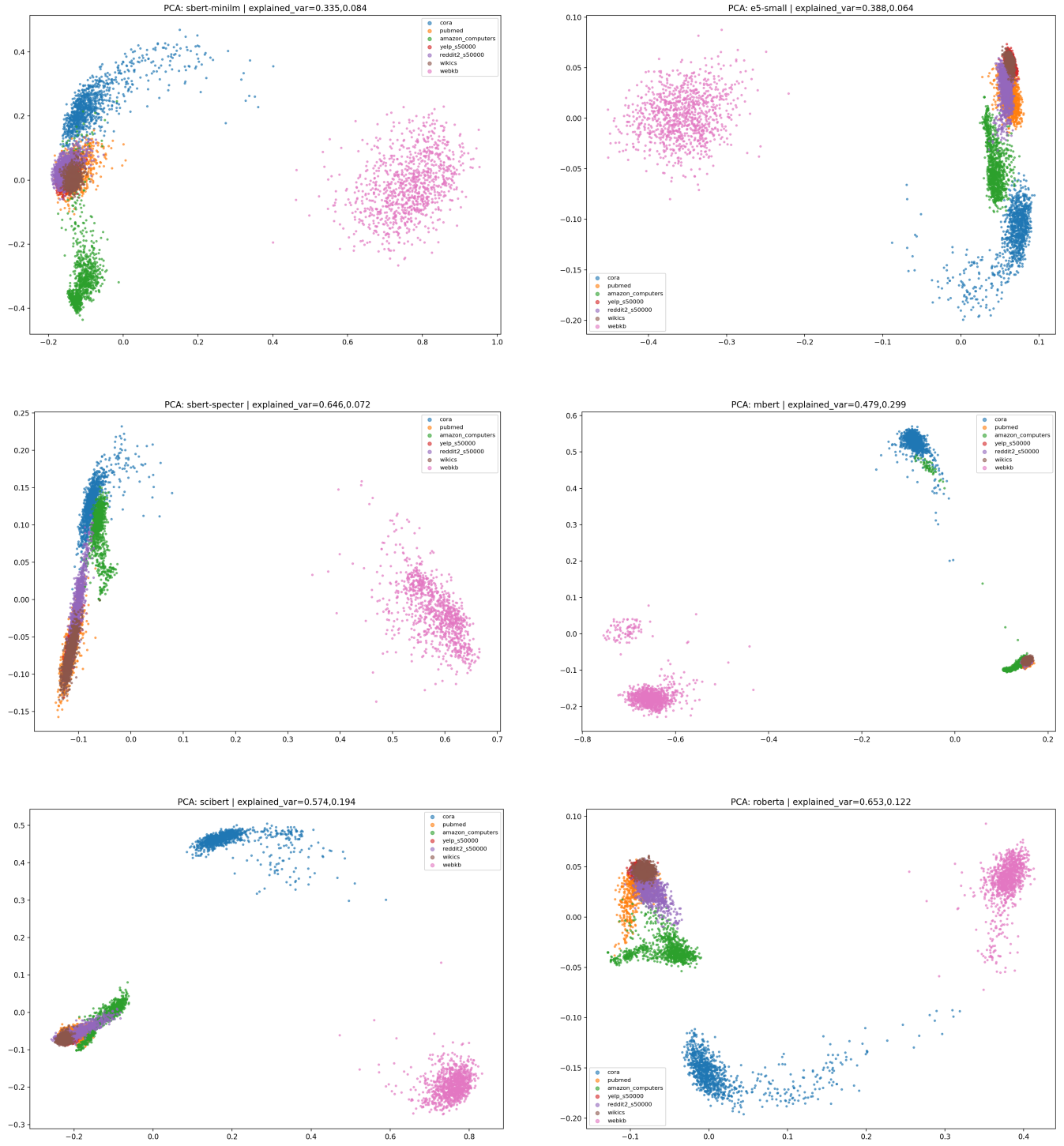


Figure 1: PCA projections of embedding spaces induced by different language models. Each subfigure shows embeddings from all datasets projected into two dimensions for a single model.

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